QUANT RESEARCH / FIXED INCOME

Bonds and RV

Understand the objects. Derive the trade.

Pricing basics are brief. Unfamiliar trading and research links receive more explanation. Every main topic answers: *what problem are we solving, what do the symbols mean, and what do we do with the answer?*

The complete chain

Market quotes → fitted fair-value curve → candidate residual → forecast of convergence → hedged position → net P&L after funding/execution.

Calibration and differentiation solve different links: calibration builds the valuation object; differentiation measures how that object's prices change when market inputs move. Neither alone proves that an RV strategy earns money.

Suggested reading route

Session	Pages	Goal
1	2–11	Connect curves, quote risk, hedge construction, carry and financing. Skim familiar derivatives; work the calibration example.
2	12–20	Understand actual bond, swap and futures trades. Explain legs, sign, carry and failures aloud.
3	21–23, 26–27	Validate research, explain portfolio units, then do drills and repair gaps. Pages 24–25 are optional breadth.

Read actively: after each page, answer its **use** or **defend** prompt. Do not memorise an equation without naming the economic object it describes.

What RV means

Buy relatively cheap exposure and hedge/sell linked expensive exposure. The link can be contractual replication or an economic/statistical relationship. Selected common risks are hedged; financing, liquidity, model and regime risks may remain.

Units throughout

N is face value; P is a dirty bond price per unit face unless labelled *per 100*. Currency value is $V = NP$, or $V = NP/100$ for a per-100 price. Lowercase p_i denotes an observed calibration price; $P_i(\theta)$ denotes its model price. Swap value per unit notional is v_{swap}, so currency value is Nv_{swap}.

$D(t) = D(0,t)$ is today's discount factor to time t; C_j is a payment in the same face/price units as P. Times are in years unless stated. Annual rates are decimals; $1\text{bp} = 10^{-4}$. When an equation uses rate moves in bp, it says so. Positive long DV01 means a long ordinary bond loses on a yield rise.

Reading the mathematical notation

Δ means a finite change; d an infinitesimal change; ∂ a partial derivative. A hat marks an estimate; $\top$ means matrix transpose; $\text{diag}(a_i)$ puts the a_i on a matrix's diagonal and zeros elsewhere. $\mathbb{E}$, Var and Cov mean expectation, variance and covariance; subscript t means conditional on information at t. All examples and graphs are hypothetical.

CORE / 02

Rates are different summaries of cash flows

Goal: compare bonds using a consistent valuation curve.

Price first; YTM second

Take one bond with n payments. t_j is payment date j in years, C_j the coupon plus any principal paid then, and $D(t_j)$ the discount factor from today. For fixed known payments,

$$P = \sum_{j=1}^{n} C_j D(t_j), \qquad P^{dirty} = P^{clean} + AI.$$

AI is accrued interest. Settlement uses dirty price. Optional/defaultable promised cash flows require the appropriate exercise/survival model.

Yield to maturity (YTM) is the single internal rate of return (IRR) reproducing P from those payments. Equal maturity and different coupons can give identical payment dates but *different payment weights*. On a non-flat curve their fair YTMs can differ; YTM is not generally the final-maturity zero rate.

Define the rates before using them

T is a future maturity in years; $z(T)$ is its continuously compounded zero rate. $f(T)$ is today's instantaneous forward rate at T. Define $D(T) = e^{-Tz(T)}$ and differentiate:

$$z(T) = -\log D(T)/T, \qquad f(T) = -\frac{\partial \log D(T)}{\partial T} = z(T) + Tz'(T).$$

Here $z'(T) = \partial z(T)/\partial T$. For dates $a < b$, let $\delta_{a,b}$ be their year fraction and $F_{a,b}$ the annual simple forward rate. Equating direct and reinvested proceeds gives $D(a)/D(b) = 1 + \delta_{a,b}F_{a,b}$.

An annuity factor converts coupon rate into value

An annuity is a stream of regular payments. Let α_j be coupon period j's year fraction and c the annual coupon rate. Coupon per unit face is $c\alpha_j$. Define its annuity factor A; present value (PV) of the coupon stream is cA:

$$A = \sum_{j} \alpha_j D(t_j), \qquad P = cA + D(T).$$

A is the *annuity factor*, with year units; principal redemption is the separate $D(T)$ term. For a par bond ($P = 1$), rearrange to $c_{par} = [1 - D(T)]/A$. Example: $c = .05$, $A = 4$ gives coupon PV .20 per unit face.

Bootstrap: solve prices for discount factors

If all earlier factors are known, isolate the final payment:

$$D(t_n) = \frac{P - \sum_{j<n} C_j D(t_j)}{C_n}.$$

Per 100: one-year zero price 96, two-year annual 5% bond price 101. Write $D_j = D(j)$:

$$D_1 = .96, \quad 101 = 5D_1 + 105D_2 \;\Rightarrow\; D_2 = .91619048.$$

Thus $z(2) = -\log D_2/2 = 4.37655\%$ and $F_{1,2} = D_1/D_2 - 1 = 4.78170\%$, using a one-year simple forward period.

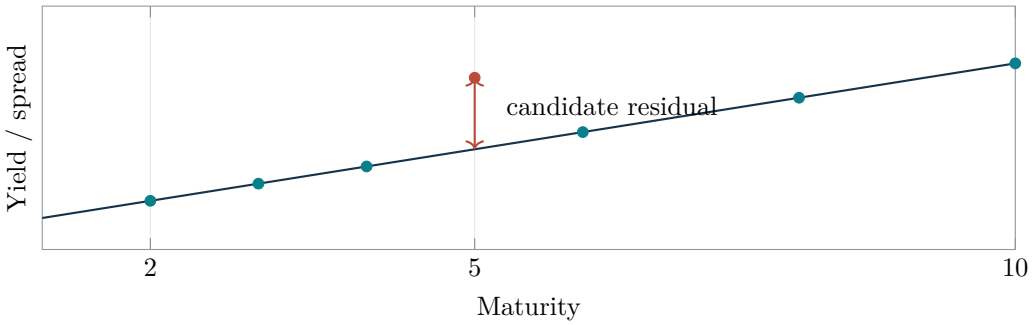
Use: reprice inputs, value compatible payments, construct forwards and calculate risk. The curve is a valuation reference, not a forecast of realised rates.

CORE / 03

Why fit a fair-value curve?

Goal: decide what a bond should cost relative to comparable instruments.

Suppose the desk observes prices of many bonds of one sovereign. They differ in coupon and maturity; some quotes are noisy or affected by scarcity. We want a common, reasonably smooth valuation reference, then ask which bonds deviate from it.



What the objects mean

There are n quoted bonds, indexed by i; j indexes a bond's payments. C_{ij} is bond i's payment at t_{ij}. Use the same price basis for every quote and model price. Write $\theta = (\theta_1, \ldots, \theta_k)^\top$ for k curve parameters.

p_i	Observed market dirty price of bond i: an input.
θ	Unknown curve parameters: e.g. zero rates at nodes or spline coefficients.
$D_\theta(t)$	Discount factor at t implied by those parameters.
$P_i(\theta)$	Model price $\sum_j C_{ij} D_\theta(t_{ij})$ for bond i's actual payments.
$\hat{\theta}$	Fitted parameters after calibrating to the quotes.
$e_i = P_i(\hat{\theta}) - p_i$	Model-minus-market price residual: positive means model-cheap.

What do we do with the result?

Use $D_{\hat{\theta}}$ to price other bonds; compare residuals across securities; calculate hedge sensitivities. A residual is only a *candidate*: it may reflect liquidity, repo, tax, quote errors or an inadequate curve model.

An exact curve through every input leaves no input residual. A fair-value fit may deliberately smooth, exclude special benchmarks or hold a candidate out. Compare alternative sensible curves before treating the residual as alpha.

Curve interpretation: identify the actual meaning of "bond funding curve." A bond-price-implied curve can absorb credit/liquidity. A repo curve measures secured cash funding. an overnight indexed swap (OIS) curve discounts under a collateral framework; a projection curve forecasts an index; internal funding is firm-specific. These objects cannot be substituted silently.

CORE / 04

What is the fitting objective doing?

Goal: choose a curve that balances quote errors and unwanted shape.

1. Define the errors and their weights

For bond $i = 1, \ldots, n$, p_i is observed dirty price and $P_i(\theta)$ its model price. Define $e_i(\theta) = P_i(\theta) - p_i$. A positive error means the model values the bond above its market price.

$w_i \geq 0$ **is the weight assigned to bond i's squared price error.** A larger w_i makes that error count more in the fit; zero excludes it. It is a modelling choice, not a bond price or portfolio holding. Define the vectors $p = (p_1, \ldots, p_n)^\top$, $P(\theta) = (P_1(\theta), \ldots, P_n(\theta))^\top$ and the matrix $W = \mathrm{diag}(w_1, \ldots, w_n)$. Then

$$\tfrac{1}{2}\sum_{i=1}^{n} w_i e_i(\theta)^2 = \tfrac{1}{2}[P(\theta) - p]^\top W[P(\theta) - p].$$

Squaring avoids cancellation; 1/2 simplifies derivatives. With errors .10 and .10 and weights 4 and 1, the first error contributes four times as much. Only relative weights matter *if the smoothness penalty below is scaled consistently.*

2. Choose what counts as a large error

Quote reliability. Let $\sigma_{quote,i} > 0$ be the estimated standard deviation of quote noise in bond i's price units. Independent Gaussian errors have negative log likelihood $\frac{1}{2}\sum_i e_i^2/\sigma_{quote,i}^2$ plus a constant. Thus $w_i = 1/\sigma_{quote,i}^2$: reliable quotes receive more weight.

Equal yield errors. Let d_i^{price} be the loss in this bond's price units for a 1bp yield rise. Let $\Delta y_{i,bp}$ be the bond's yield change in basis points. Then $\Delta P_i \approx -d_i^{price}\Delta y_{i,bp}$, so yield-equivalent error is $-e_i/d_i^{price}$. Minimising its squared size uses $w_i = 1/(d_i^{price})^2$. This is not automatically a quote-noise model.

> Per 100, a .10 price error and price DV01 .02 correspond to 5bp; the same error with price DV01 .10 corresponds to 1bp. Equal price weights treat them equally; equal yield-error weights do not. Dollar DV01 cannot be substituted for price DV01 without converting units.

3. Define the smoothness penalty

L is a chosen difference matrix acting on the k parameters; $\mathsf{R} = L^\top L$ is its $k \times k$ penalty matrix. For three equally spaced rate nodes, $L = (1, -2, 1)$ makes $\|L\theta\|^2 = (\theta_1 - 2\theta_2 + \theta_3)^2$: a squared local bend. $\|\cdot\|$ is Euclidean length. $\lambda_{sm} \geq 0$ sets penalty strength in matching objective units.

$$\hat{\theta} = \arg\min_{\theta} \left\{ \tfrac{1}{2}[P(\theta) - p]^\top W[P(\theta) - p] + \tfrac{\lambda_{sm}}{2}\theta^\top \mathsf{R}\theta \right\}.$$

arg min returns the parameter vector attaining the minimum. Large λ_{sm} favours smoothness; tiny λ_{sm} can fit noise. Treat W, L, λ_{sm} as fixed in the differentiation on the next general chain-rule page.

Use: the output $\hat{\theta}$ gives fair prices and candidate residuals. Validate errors, forwards, stability and held-out bonds; smoothness alone establishes neither no-arbitrage nor profitable convergence.

CORE / 05

Quote risk: start with a bootstrap

Goal: see how changing an input quote changes a different bond's value.

The input prices determine both curve nodes

All prices here are **per 100 face**. Keep the notation p_i for market input prices. Let p_1 price a one-year zero redeeming 100; let p_2 price a two-year annual 5% coupon bond redeeming 100. Set $D_1 = D(1)$ and $D_2 = D(2)$.

$$p_1 = 100D_1, \qquad p_2 = 5D_1 + 105D_2.$$

Rearrange in the order of the bootstrap:

$$D_1 = p_1/100, \qquad D_2 = (p_2 - 5D_1)/105.$$

For another two-year bond paying 3 at year 1 and 103 at year 2, call its model price P_*:

$$P_* = 3D_1 + 103D_2 = 3\frac{p_1}{100} + \frac{103}{105}\left(p_2 - 5\frac{p_1}{100}\right).$$

The star only identifies this target bond. Unlike a currency position value, P_* is still a price per 100.

Differentiate one quote at a time

Hold p_1 fixed and change p_2. The first node does not change; the second changes by $\partial D_2/\partial p_2 = 1/105$. Therefore

$$\frac{\partial P_*}{\partial p_2} = 103\frac{1}{105} = .980952.$$

Now hold p_2 fixed and change p_1. Both nodes move:

$$\frac{\partial D_1}{\partial p_1} = \frac{1}{100}, \qquad \frac{\partial D_2}{\partial p_1} = -\frac{5}{105}\frac{1}{100}.$$

Apply the two sensitivities to the target's payments:

$$\boxed{\frac{\partial P_*}{\partial p_1} = 3\frac{1}{100} + 103\left(-\frac{5}{10500}\right) = -.0190476.}$$

The negative sign is economically meaningful: a higher first quote raises D_1, but D_2 must fall to preserve the second quote. The target has relatively more weight at year 2, so its net value falls. Freezing D_2 would incorrectly give $+.03$.

At $p_1 = 96$, $p_2 = 101$, $P_* = 97.247619$. A .10 increase in p_2 raises P_* by .0980952; a .10 increase in p_1 lowers it by .00190476. These relations are exact here because the bootstrap and target price are linear in the quotes.

How to use this for a hedge

For small quote changes,

$$\Delta P_* \approx -.0190476\Delta p_1 + .980952\Delta p_2.$$

Long one target face unit can be hedged against these quote moves by *long* .0190476 face units of bond 1 and *short* .980952 face units of bond 2. With matched face/price conventions, their combined payments cancel the target's payments exactly in this ideal example. General curve fits need not yield an exact replication.

Use: translate the model valuation into observable quote exposures and hedge holdings. Multiply a per-100 price change by $N/100$ to obtain currency P&L for face N.

CORE / 06

The calibration chain rule

Goal: generalise the bootstrap calculation without hiding what its matrices mean.

Define the calibration map

$p \in \mathbb{R}^n$ is the vector of n market input prices; $\theta \in \mathbb{R}^k$ contains k curve parameters. Calibration selects $\hat{\theta}(p)$ through k equations $g(\theta, p) = 0$. Each component of g is an equation error, not a bond price.

For an exact square bootstrap, $k = n$ and one can take $g = P(\theta) - p$. For page 4's weighted fit, g is instead the gradient of its objective. Define the $n \times k$ price Jacobian $J = \partial P/\partial\theta$, whose entry $J_{ij} = \partial P_i/\partial\theta_j$. With fixed $W, \mathsf{R}, \lambda_{sm}$,

$$g(\theta, p) = J^\top W[P(\theta) - p] + \lambda_{sm}\mathsf{R}\theta = 0.$$

This follows by differentiating the squared-error objective with respect to each parameter. It is the first-order condition for an interior optimum.

Differentiate equations that must remain satisfied

Define $G_\theta = \partial g/\partial\theta$ $(k \times k)$ and $G_p = \partial g/\partial p$ $(k \times n)$, evaluated at the calibrated solution. A small input change must satisfy

$$0 = \mathrm{d}g = G_\theta \mathrm{d}\theta + G_p \mathrm{d}p.$$

Let $B_\theta = \partial\hat{\theta}/\partial p$ $(k \times n)$ be the parameter response to quotes. Substitute $\mathrm{d}\theta = B_\theta \mathrm{d}p$ and solve:

$$G_\theta B_\theta = -G_p \quad \Longrightarrow \quad B_\theta = -G_\theta^{-1} G_p.$$

The inverse is mathematical notation; calculate B_θ with a linear solver. It requires a differentiable local calibration and invertible G_θ. In the least-squares case, G_θ is the objective's Hessian (matrix of second derivatives), including residual-dependent terms for nonlinear prices.

Propagate that response into another instrument

$P_*(\theta, p)$ is a target model price in stated price units. Define row vectors $P_{*,\theta} = \partial P_*/\partial\theta$ $(1 \times k)$ and $P_{*,p} = \partial P_*/\partial p$ $(1 \times n)$, each holding the other argument fixed. The ordinary chain rule gives

$$\mathrm{d}P_* = P_{*,\theta}\mathrm{d}\theta + P_{*,p}\mathrm{d}p = [P_{*,p} + P_{*,\theta}B_\theta]\,\mathrm{d}p.$$

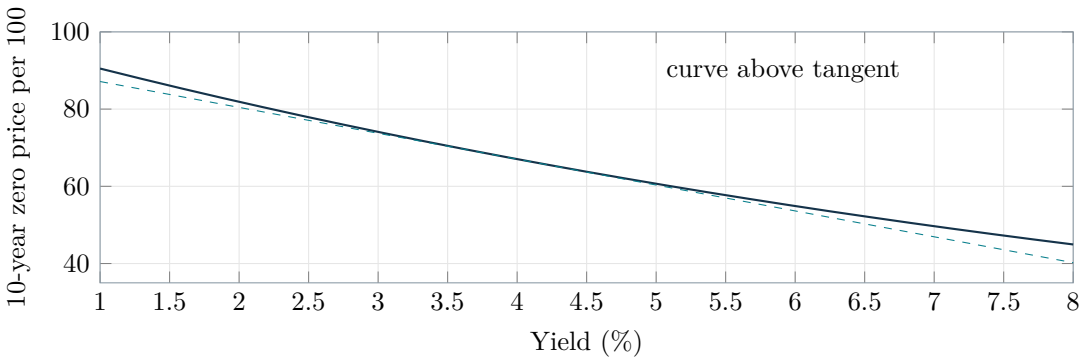
Thus total quote sensitivity is the direct term plus the recalibration term. The direct term is zero when quotes affect the target only through the curve. Holding the curve frozen discards the recalibration term.

Use: obtain quote hedges and explain daily price moves. Compare analytic results against quote bumps followed by full recalibration. A nearly singular G_θ can amplify quote noise and destabilise risks; it does not establish a trading edge.

CORE / 07

Duration, DV01 and convexity

Goal: turn a rate move into approximate currency P&L and hedge quantities.



Let y be annual YTM compounded m times/year; t_i are payment times in years and C_i the payments in the price's face units. $P\prime$ and $P\prime\prime$ mean first and second derivatives with respect to y. For this fixed cash-flow schedule,

$$P(y) = \sum_i C_i(1+y/m)^{-mt_i},$$
$$P'(y) = -\sum_i t_i C_i(1+y/m)^{-mt_i-1},$$
$$P''(y) = \sum_i t_i(t_i+1/m)C_i(1+y/m)^{-mt_i-2}.$$

$D_{mod} = -P'/P$ is modified duration (years); $\mathcal{C} = P''/P$ is convexity (years squared). V is the position's currency market value and Δy a yield change in decimal units. Taylor expansion gives

$$\Delta V \approx -V D_{mod}\Delta y + \tfrac{1}{2}V\mathcal{C}(\Delta y)^2, \qquad \text{DV01} = V D_{mod} 10^{-4}.$$

DV01 is the first-order loss of a long ordinary bond for +1bp. Convexity captures curvature of the price–yield relationship; the graph shows why a tangent misses it.

$V = £10m$, $D_{mod} = 7$, $\mathcal{C} = 60$, shock +10bp = .001:

$$\text{DV01} = £7,000/\text{bp}, \qquad \Delta V \approx -70,000 + 300 = -£69,700.$$

Use market value, not face, in this currency formula. Optionality can change convexity's sign.

Define $\omega_i = C_i(1+y/m)^{-mt_i}/P$, the fraction of price contributed by payment i. Macaulay duration, the PV-weighted payment time, is $D_{Mac} = \sum_i \omega_i t_i$, and $D_{mod} = D_{Mac}/(1+y/m)$. A YTM bump, a parallel-zero bump and a key-rate bump are different scenarios; label the sensitivity you use.

Use: estimate local P&L, size hedges and identify exposures left after a hedge. Large shocks or options may require full repricing.

CORE / 08

Construct a factor hedge

Goal: keep the intended relative-value exposure while reducing unwanted rate factors.

Suppose a target bond is cheap but also loses if all rates rise. Hedge the common rate move so the trade depends more on its relative pricing. Matching face amounts generally fails because bonds have different sensitivities.

Define the hedge system

$a \in \mathbb{R}^k$: target position's signed sensitivities to k chosen factors. $\mathsf{H} \in \mathbb{R}^{k \times m}$: sensitivities per unit of m hedge instruments. $\mathbf{h} \in \mathbb{R}^m$: signed holdings of the m hedge instruments (negative means short). Residual exposure is $a + \mathsf{H}\mathbf{h}$; exact neutrality requires

$$\boxed{\mathsf{H}\mathbf{h} = -a.}$$

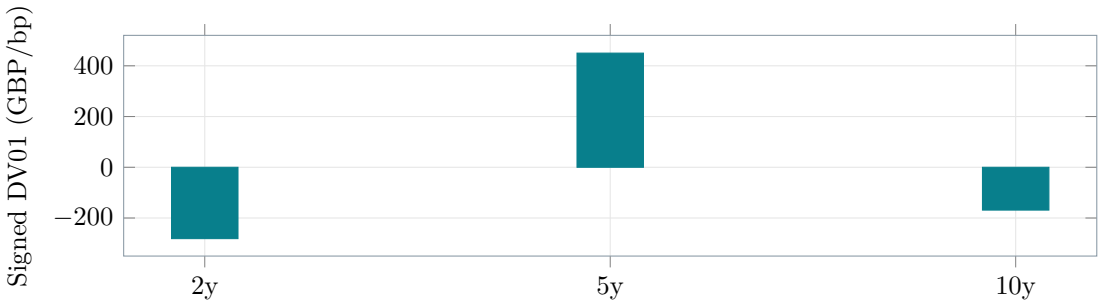
k is the number of factors, m the number of available hedges, and $\mathbb{R}^k$ denotes a real k-component vector. Exposures must use the same factor units. This is exposure matching, not a forecast. Specify factors, units and signs first.

Worked level-and-slope butterfly

Per £1m face, 2y/5y/10y long DV01s are 200/450/800. Long 1m of 5y. Approximate slope loading by maturity times DV01. Signed wing notionals u_S, u_L in millions solve

$$\begin{aligned} 200u_S + 450 + 800u_L &= 0, \\ 2(200u_S) + 5(450) + 10(800u_L) &= 0. \end{aligned}$$

Subtract twice the first equation: $1350 + 6400u_L = 0$, hence $u_L = -.2109375$. Then $u_S = (-450 - 800u_L)/200 = -1.40625$.



Signed DV01s sum to zero. The notionals do not. The slope constraint is also zero under the chosen approximate loading.

Use: a belly-only 3bp yield fall earns about $450(3) = £1,350$ before costs. A different twist, convexity, repo or model change can still hurt. Real factors should reflect calibrated or empirical exposures, not maturity alone.

CORE / 09

Which hedge are we trying to optimise?

Goal: distinguish structural neutrality from low historical P&L variance.

A DV01 hedge matches a specified shock

The system on page 8 makes selected factor sensitivities zero. It does not necessarily minimise the trade's observed P&L variance: factors move together, and instruments have other basis/liquidity risks.

A regression hedge minimises empirical variance

Let Π_A be target P&L and Π_B one hedge unit's P&L over the same horizon; η is the number of hedge units sold. Write the hedged outcome as $\Pi_A - \eta\Pi_B$. Expand:

$$\begin{aligned}\operatorname{Var}(\Pi_A - \eta\Pi_B) &= \operatorname{Var}\Pi_A - 2\eta\operatorname{Cov}(\Pi_A,\Pi_B) + \eta^2\operatorname{Var}\Pi_B,\\ \partial_\eta \operatorname{Var} &= -2\operatorname{Cov}(\Pi_A,\Pi_B) + 2\eta\operatorname{Var}\Pi_B = 0,\\ \eta^* &= \operatorname{Cov}(\Pi_A,\Pi_B)/\operatorname{Var}\Pi_B.\end{aligned}$$

If correlation is .8, target volatility 10 and hedge-unit volatility 4, then $\eta^* = .8(10)/4 = 2$. That ratio depends on the estimated distribution. It can leave nonzero DV01 under your chosen structural shock.

PCA chooses the common risk directions

Let Δy be a vector of centred historical tenor-yield changes and $\Sigma_y = \operatorname{Cov}(\Delta y)$ its covariance matrix. Principal component analysis (PCA) decomposes it as $\Sigma_y = U\Lambda U^\top$. Columns u_j of U are orthonormal eigenvectors; diagonal entries ℓ_j of $\Lambda = \operatorname{diag}(\ell_j)$ are their variances (eigenvalues). Factor score j is $u_j^\top \Delta y$. Large ℓ_j identify directions explaining more variance. Level/slope/curvature are common interpretations, not guaranteed mathematical labels. Eigenvector signs are arbitrary.

Covariance-PCA emphasises volatile tenors; correlation-PCA standardises them first. Estimate loadings on past data, then hedge or measure exposure to those loadings. A hedge removes exposure to the chosen *risk description*; PCA itself predicts no convergence.

When an exact solution is unstable

If two hedges have nearly identical sensitivities, exact constraints can demand large offsetting positions. Use page 8's target exposures a, hedge matrix H and holdings $\mathbf{h}$. Define Ω_f as a positive-semidefinite $k \times k$ factor-risk weighting matrix (e.g. factor-shock covariance), R_h as a positive-semidefinite $m \times m$ holdings penalty, and $\lambda_h \geq 0$ as its strength. Minimise

$$(a + \mathsf{H}\mathbf{h})^\top \Omega_f (a + \mathsf{H}\mathbf{h}) + \lambda_h \mathbf{h}^\top \mathsf{R}_h \mathbf{h}.$$

Differentiate with respect to $\mathbf{h}$, cancel the common factor 2 and set the gradient to zero:

$$(\mathsf{H}^\top \Omega_f \mathsf{H} + \lambda_h \mathsf{R}_h)\mathbf{h} = -\mathsf{H}^\top \Omega_f a.$$

Use: choose a hedge consistent with the objective, report residual risks and stress unstable estimates. Dollar-, DV01-, regression- and PCA-neutrality are different statements. None makes financing risk disappear.

CORE / 10

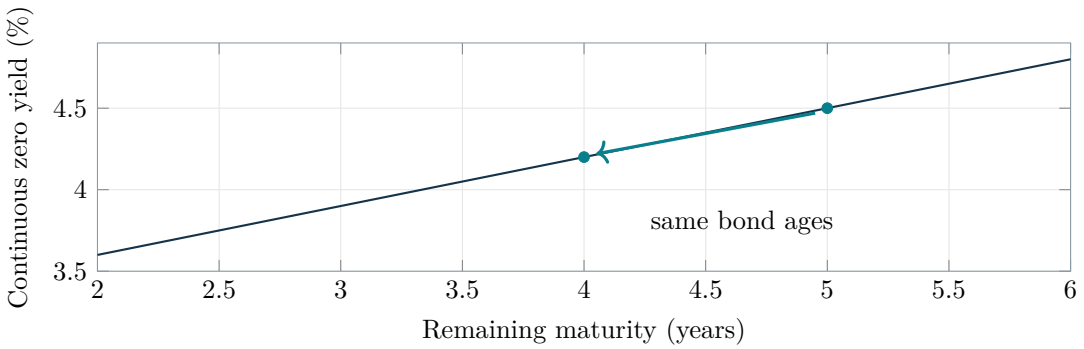
Carry, rolldown and pull-to-par

Goal: explain holding-period P&L even before forecasting a market move.

Let h be the holding horizon, N unchanged face value, P_0^{dirty}, P_h^{dirty} prices per unit face, and C_i bond payments per unit face paid at t_i (including redemption if within the horizon). Net currency P&L Π_h is:

$$\Pi_h = N\left[P_h^{dirty} - P_0^{dirty} + \sum_{0<t_i\le h} C_i\right] - \text{funding} - \text{costs} + \Pi_{hedge}.$$

Π_{hedge} is the hedge legs' net P&L; funding and costs are currency amounts for the same period, with no double counting across legs. This is wealth accounting. Carry/roll are scenario attributions of that wealth; conventions differ. Accrued is already inside dirty prices. Include coupon reinvestment if relevant.



Separate two effects with a zero, per 100

Pay 100 in five years. Today's zero curve has $z(5) = 4.5\%$, $z(4) = 4.2\%$. At a one-year horizon:

$$P_0 = 100e^{-.045(5)} = 79.85162,$$
$$P_{1,\textit{same yield}} = 100e^{-.045(4)} = 83.52702,$$
$$P_{1,\textit{static curve}} = 100e^{-.042(4)} = 84.53538.$$

First age at the old yield: +3.67540. Then reprice at the curve's shorter-maturity yield: another +1.00836. Total gross return 5.86558%, before financing/costs. No coupon is needed for ageing/accretion to create return.

Carry: holding/accrual/accretion net of financing under a stated baseline.
Rolldown: same bond ages along an assumed unchanged maturity curve.
Pull-to-par: discount/premium amortises under a fixed-yield baseline.

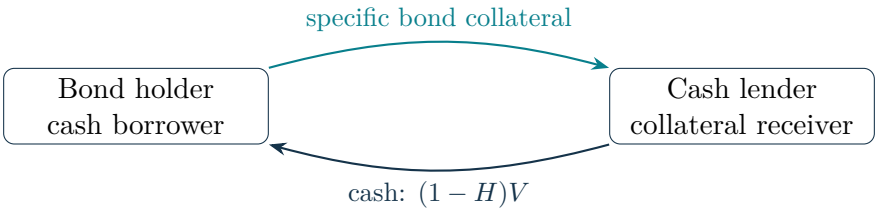
Use: compare *net* holding economics across candidate trades, including short legs. A static maturity curve uses $D(T-h)$ at horizon; today's forwards being realised uses $D(T)/D(h)$, with T the bond's original remaining maturity and D today's discount function. These are different assumptions, neither a guarantee. Rolldown can hurt on an inverted local curve.

CORE / 11

Repo and specialness: follow the flows

Goal: find the actual financing/borrow cost of holding an RV position.

Here V is collateral market value, $H \in [0, 1)$ its haircut fraction, and τ the financing year fraction.



A bond holder supplies collateral and borrows cash; at maturity it repays cash plus repo interest and receives the bond back. This is repo from its perspective, reverse repo from the cash lender's. A short can source the specific required security by lending cash against it, then selling it, subject to actual borrow/settlement terms.

Why scarce collateral is cheap to finance, costly to borrow

General collateral (GC) repo is cash funding against eligible collateral. Denote its annual rate by r_{GC}, a specific issue's annual rate by $r_{special}$, and the actual repo rate by r_{repo}. A scarce issue can trade at $r_{special} < r_{GC}$:

- Its holder can post it and pay a lower cash borrowing rate.
- The trader sourcing it lends cash at a lower return than available on GC collateral.

The second trader gives up approximately $V(r_{GC} - r_{special})\tau$. The sign follows from whose cash is lent and whose bond is needed.

£10m, GC 4.5%, special 1%, 30/360:

$$10^7(.045 - .01)(30/360) = £29,166.67.$$

An expected £10,000 convergence gain does not cover that sourcing opportunity cost under these assumptions.

Haircuts and short carry

Haircut H means repo cash $(1 - H)V$, so HV needs equity or another funding source. If the annual equity/alternative funding cost is r_e, approximate financing cost per year is $V[(1 - H)r_{repo} + Hr_e]$. Fees and margin changes add further terms.

A short has coupon-equivalent obligations and borrow/rebate terms. Sale proceeds may be encumbered. Do not assume they freely finance the long at OIS.

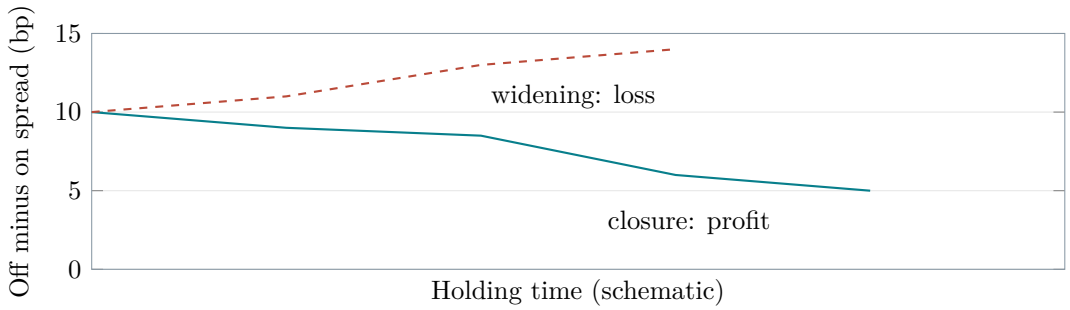
Use: evaluate each leg's holding cash flows and funding availability. Specific-collateral financing can explain why an apparently rich bond remains rich. A spread model that ignores it can mistake collateral value for alpha.

TRADES / 12

On-the-run versus off-the-run

Goal: distinguish a fair liquidity premium from a convergence opportunity.

The on-the-run bond is the latest issue in a benchmark tenor; older issues are off-the-run. Benchmark liquidity and collateral usefulness can make the on-the-run issue expensive. Coupon, maturity and risk differences also affect observed yields.



Define the spread before taking a position

y_{off} and y_{on} are the two bond yields expressed in bp. Define their gap as $s_{off/on} = y_{off} - y_{on}$. $\Delta\Pi$ denotes currency trading P&L. For a long off / short on trade with matched dollar DV01 d (currency loss per +1bp on each long leg),

$$\Delta\Pi \approx -d\Delta y_{off} + d\Delta y_{on} = -d\Delta s_{off/on}.$$

The trade gains when the spread narrows. With $d = £3,000$/bp and $s_{off/on} : 12 \to 8$bp, gross first-order gain is $+£12,000$. A widening to 19bp loses £21,000. Funding, coupon and execution terms must then be included.

What should converge?

A raw yield gap need not converge to zero. Fit coupon/maturity effects and consider issue age, liquidity, repo specialness and auction status. If conditional fair spread is 7bp, a 12bp observation is 5bp wide relative to that model. The model may still be wrong.

Shorting the rich benchmark may require scarce collateral. Its sourcing cost can absorb the apparent yield advantage; specialness can worsen as the trade loses. New issuance can change benchmark status and liquidity rather than produce immediate convergence.

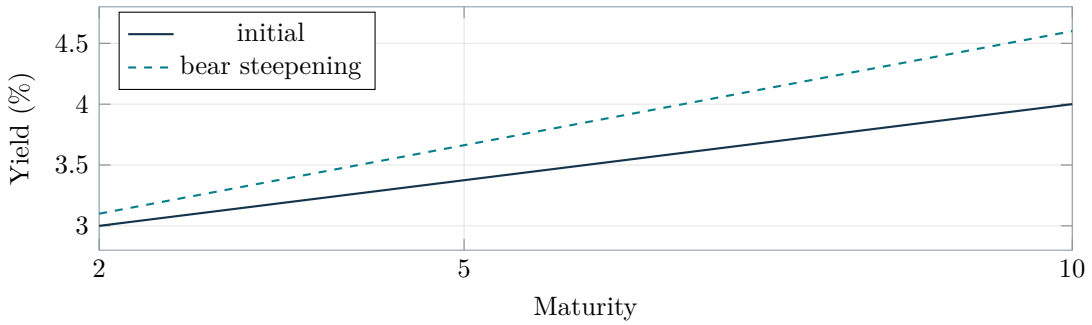
Use: identify a residual beyond explainable differences, forecast its change over a stated horizon, and compare its expected dollar closure with both legs' net carry and costs. Hedge key-rate differences as well as total DV01.

Defend: why would this premium change, and what observation would tell you your original thesis was false?

TRADES / 13

Slope, curvature and issuance RV

Goal: isolate a curve shape or local dislocation instead of an outright rate view.



A steepener is a signed exposure

Write y_S, y_M, y_L for short-, medium- and long-maturity yields in bp. Define slope $s_{slope} = y_L - y_S$; $d > 0$ is the matched currency DV01 magnitude and $\Delta\Pi$ is currency P&L. Long the shorter bond and short the longer bond, each with DV01 d, gives

$$\Delta\Pi = -d\Delta y_S + d\Delta y_L = d\Delta s_{slope}.$$

It gains if slope rises, including a bear steepening where both yields rise but the long yield rises more. A flattener reverses the legs. Matching DV01 removes one parallel-shift exposure, not all curve risk.

A butterfly asks whether the belly is unusual

Define a yield butterfly $b_{fly} = y_M - a_{fly}y_S - (1 - a_{fly})y_L$, where a_{fly} sets the short-wing share of total wing DV01. The choice of a_{fly} defines the comparison: interpolation in maturity, fitted factor loadings or a statistical hedge. Long belly with signed wing DV01s $-a_{fly}d$ and $-(1 - a_{fly})d$ gives

$$\Delta\Pi \approx -d[\Delta y_M - a_{fly}\Delta y_S - (1 - a_{fly})\Delta y_L] = -d\Delta b_{fly}.$$

This buys a belly whose relative yield is high, expecting b_{fly} to fall. Dollar-risk weights become notionals only after division by each instrument's DV01. Page 8 constructs a level-and-slope-neutral example.

Issuance, auctions and strips

New supply, dealer inventory and client demand can temporarily alter a sector's relative price. An auction-concession hypothesis predicts cheapening into supply and reversal afterwards; neither timing nor reversal is assured. Test announcement, auction and settlement times separately.

A coupon bond is a basket of dated payments. Where stripping/reconstitution is available, compare its price with the cost of matching coupon and principal strips. The replication identifies an economic anchor, but executable bid/ask, financing, processing and settlement determine whether it is tradable.

Use: name the precise shape, the hedge weights, the event clock and the cash-flow link. A smooth-looking curve is not evidence that every local deviation is mispricing.

TRADES / 14

Swaps: the hedge and the benchmark

Goal: separate a bond-specific spread view from broad interest-rate exposure.

Build the par swap rate from its cash flows

Consider unit notional and a swap starting today, with final maturity T. Payment j is at t_j, accrual fraction α_j is in years, and $D_j = D(t_j)$. The fixed-leg annuity factor is

$$A = \sum_j \alpha_j D_j.$$

It is the PV of receiving one unit of fixed rate per year. In an ideal single-curve setting, a floating-rate note resets to par; its floating coupons have PV $1 - D(T)$. Let K_c be the contractual annual fixed rate. Receive-fixed/pay-floating swap value per unit notional is v_{rec}, while $v_{pay} = -v_{rec}$. Therefore

$$v_{rec} = K_c A - [1 - D(T)].$$

Let K_{par} be today's par fixed rate (making a new swap worth zero). Setting $v_{rec} = 0$ gives $K_{par} = [1 - D(T)]/A$. We have converted a discount curve into a comparable quoted rate.

Why paying fixed hedges a long bond

A long fixed-rate bond loses when rates rise. An existing pay-fixed swap gains: its contractual payments are cheap relative to the newly available higher fixed rate. Since $v_{pay} = A(K_{par} - K_c)$, differentiation at par ($K_{par} = K_c$) gives $\mathrm{d}v_{pay} = A\mathrm{d}K_{par}$; the term $(K_{par} - K_c)\mathrm{d}A$ vanishes. Rate DV01 magnitude is therefore approximately $NA10^{-4}$ for notional N.

Let bond and pay-fixed swap have matched dollar DV01 d. For bond yield y_b and par swap rate K_{par}, with both changes expressed in bp,

$$\Delta\Pi \approx -d\Delta y_b + d\Delta K_{par} = d\Delta(K_{par} - y_b).$$

The hedged position gains if the bond becomes richer relative to swaps. Maturity matching alone does not establish equal key-rate exposure.

What changes with multiple curves?

Let F_j be the projected annual floating rate for accrual period j, and PV_{float} the floating coupons' PV per unit notional. Use the collateral-consistent curve for discounting and the index-specific curve for projection:

$$PV_{float} = \sum_j D_j \alpha_j F_j, \qquad K_{par} = PV_{float}/A.$$

The shortcut $PV_{float} = 1 - D(T)$ requires the single-curve assumptions above. Basis risk between discounting and projection remains.

Use: convert a bond position into bond-versus-swap spread exposure. A rate hedge does not remove issuer default, liquidity, financing or contractual mismatch. Calculate sensitivities to each relevant curve quote.

TRADES / 15

Asset swaps: why convert the coupon?

Goal: compare bond cheapness as a floating-rate spread, including price away from par.

An asset-swap package combines a fixed-rate bond with a swap paying its coupon and receiving floating plus spread s_{ASW}. This turns the bond's fixed coupon into a floating receipt. A package convention determines how the purchase premium/discount is funded or exchanged.

Derive one explicit par-package convention

Use unit face, matching coupon/swap dates and fractions, no default, a single discount/projection curve, and no accrued-interest complications. Let c be the bond annual coupon rate, P_b its market price per unit face, and $D = D(T)$ its maturity discount factor. The annuity factor is $A = \sum_j \alpha_j D(t_j)$, with α_j the coupon accrual year fraction at payment time t_j; $K_{par} = (1 - D)/A$ is the par swap rate. Write B_{ref} for its reference-curve price per unit face:

$$B_{ref} = cA + D.$$

The swap receives floating plus annual spread s_{ASW} and pays c. Its PV per unit notional is v_{swap}:

$$v_{swap} = 1 - D + s_{ASW}A - cA = 1 - B_{ref} + s_{ASW}A.$$

Require the initial bond-plus-swap package value to equal par:

$$1 = P_b + v_{swap} \quad \Longrightarrow \quad \boxed{s_{ASW} = \frac{B_{ref} - P_b}{A} = c - K_{par} + \frac{1 - P_b}{A}.}$$

The first term compares coupon with swap par rate. The final term converts the bond discount/premium into annual spread units. A price gap becomes a rate because A is PV per unit of annual rate.

$A = 4.5$, $D = .8$, $c = .05$, $P_b = .99$: $B_{ref} = 1.025$, $K_{par} = .2/4.5 = 4.4444\%$.

$$s_{ASW} = (1.025 - .99)/4.5 = .0077778 = 77.78\text{bp}.$$

The swap is worth $1 - 1.025 + .0077778(4.5) = .01$; bond .99 plus swap .01 equals 1. The equality checks the sign and the upfront convention.

How we use the number

A larger s_{ASW} means a larger reference-price discount per unit annuity under this convention. Compare issues, forecast spread changes, then include funding and residual risks. A swap exchanging coupons does not guarantee recovery of bond principal after issuer default.

Keep quote definitions separate: yield spread compares scalar yields; Z-spread shifts discount rates to reproduce price; asset-swap spread prices a specified package; OAS also models embedded options. Actual asset-swap schedules, upfronts, curves and default/termination provisions can differ from this teaching convention.

TRADES / 16

Financed bond forwards

Goal: find the forward price consistent with buying and financing the bond today.

Construct the replicating trade

Buy the bond at dirty price P_0 and borrow its purchase price until delivery time T. Interim coupons reduce the net amount that must be recovered at delivery: either use them to repay funding or reinvest them to T. Assume the same locked simple annual rate r for both flows.

Write $P_{0,T}^{fwd,dirty}$ for today's contracted dirty delivery price, in the same face units as P_0 and coupons C_j. Let τ be the funding year fraction to T, and τ_j the year fraction from coupon date $t_j < T$ to T. Terminal funding owed less accumulated coupons is

$$\boxed{P_{0,T}^{fwd,dirty} = P_0(1 + r\tau) - \sum_{t_j<T} C_j(1 + r\tau_j).}$$

Selling the bond forward at this price exactly covers that amount. This is the cash-and-carry replication: the forward delivery receives the same bond that was financed. Coupon entitlement at the delivery date must follow the actual settlement convention.

Dirty purchase price 100; six-month funding at 4%; coupon 1 received three months before delivery:

$$P^{fwd,dirty} = 100(1 + .04 \times .5) - 1(1 + .04 \times .25) = 100.99.$$

The accrued-interest amount at delivery converts this to the corresponding clean forward price. Adding financing without subtracting the coupon would overstate the required delivery price.

Why OIS and repo forwards can differ

Under a common frictionless discount curve, the same reasoning yields

$$P_{0,T}^{fwd,dirty} = \frac{P_0 - \sum_{t_j<T} C_j D(0,t_j)}{D(0,T)}.$$

Specific repo, borrowing constraints, haircuts, reinvestment rates and settlement costs can invalidate that common-curve replication. A special bond's financing advantage can therefore enter its forward economics.

From formula to trading decision

Compare the executable forward price with your financed replication using actual terms. To exploit an expensive forward, buy/finance the cash bond and sell the forward; to exploit a cheap forward, the reverse requires the bond to be borrowable. A theoretical inequality is not an arbitrage if financing cannot be locked or the short cannot be delivered.

Use: price delivery trades, isolate net carry from basis and understand the cash flows behind futures implied repo. State whether the financing rate is locked or merely forecast.

TRADES / 17

Treasury futures: what is delivered?

Goal: connect a standardised futures quote to the particular cash bond and its risks.

A Treasury futures contract permits delivery of an eligible bond basket. The short generally chooses the security and delivery timing within the contract rules. Each eligible bond i has conversion factor CF_i, translating the common futures quote F_{fut} into its clean invoice amount. Write I_i^{dirty} for the dirty delivery invoice per 100 face, $AI_i(T)$ for accrued interest per 100 at delivery time T, and P_i^{clean} for today's clean cash price per 100. In matching price units,

$$I_i^{dirty} = CF_i F_{fut} + AI_i(T).$$

Here $AI_i(T)$ is accrued interest at delivery. Futures contract face must also be included when converting price units into dollars.

Cheapest-to-deliver is a financing comparison

For each eligible bond, buy it today, finance it, collect coupons, then deliver it for its invoice. Compare net delivery economics. The cheapest-to-deliver (CTD) minimises the appropriately financed delivery cost relative to invoice; under aligned assumptions it maximises implied repo.

> Ignoring carry for illustration, bond A costs clean 100 with $CF = .95$; B costs 98 with $CF = .92$. Their converted prices are 105.263 and 106.522. A is cheaper on this comparison even though its unconverted cash price is higher. Actual CTD selection adds carry and delivery details.

Basis and net basis

b_i is gross clean basis in price points per 100, defined as

$$b_i = P_i^{clean} - CF_i F_{fut}.$$

Long cash / short CF_i units of appropriately scaled futures has mark-to-market change approximately Δb_i. If this price basis rises, that position gains. Do not import the opposite sign from a yield-spread convention.

Net basis adjusts gross basis for financing and coupon carry under a specified convention. Delivery relates each candidate's invoice to futures; it does not force every eligible issue to have zero basis, since the short owns a delivery choice.

Why the hedge can change

Rates, curve shape, repo and calendar changes can switch CTD. Then a contract's effective duration and hedge ratio can change. Delivery timing and quality options also affect value; treating futures as one fixed cash bond misses those options. Daily variation margin introduces cash-flow timing differences from a forward.

Use: before a cash–futures RV trade, identify CTD alternatives, delivery assumptions, basis definition, financing and the contract's current risk.

TRADES / 18

Implied repo and futures hedge size

Goal: turn delivery cash flows into a financing rate and then hedge the rate risk.

Derive implied repo instead of memorising it

Implied repo asks: which simple annual financing rate r makes buying this bond and delivering it against futures break even? Let dirty purchase price be P, dirty invoice I, and interim coupons C_j. τ is the year fraction from purchase to delivery, and τ_j from interim coupon j to delivery. P, I, C_j must use matching price units. r_{IRR} denotes the break-even annual rate called implied repo. Thus

$$0 = I + \sum_j C_j(1 + r\tau_j) - P(1 + r\tau).$$

Collect constant and r terms:

$$\boxed{r_{IRR} = \frac{I + \sum_j C_j - P}{P\tau - \sum_j C_j\tau_j}.}$$

The denominator is net time-weighted cash funding; the numerator is the gross cash surplus before interest. Units are annual rate, not a bond yield.

$P = 99.5$, $I = 99.3$, coupon 1, $\tau = .25$, $\tau_1 = .1$:

$$r_{IRR} = \frac{99.3 + 1 - 99.5}{99.5(.25) - 1(.1)} = \frac{.8}{24.775} = 3.22906\%.$$

If locked all-in actual repo is lower, the simplified financed delivery trade has positive surplus; include costs, margins and options before interpreting this as attainable profit.

Differentiate the cash–futures link

Here CF is the CTD's conversion factor and F_{fut} the futures quote; $P_{CTD}^{fwd,clean}$ is its clean forward price with accrued handled separately. If CTD stays fixed and carry effects are locally small, $P_{CTD}^{fwd,clean} \approx CF\, F_{fut}$. Differentiate:

$$DV01_{fut} \approx DV01_{CTD}^{forward}/CF.$$

$DV01_{fut}$ is currency/bp per contract; $DV01_{CTD}^{forward}$ is currency/bp for the CTD forward on one contract's face. The numerator must use the cash face represented by one futures contract. We divide by CF because a one-unit futures price move produces only CF units of converted invoice change.

If CTD forward DV01 is 72 dollars/bp per contract face and $CF = .9$, futures DV01 is 80 dollars/bp per contract. A long cash book with DV01 \$16,000/bp needs approximately 200 short contracts for a scalar rate hedge.

Use: IRR ranks delivery financing economics; DV01 sizes exposure. A hedge selected by equal notional can be materially wrong. CTD switches, key-rate mismatch and basis moves remain after scalar DV01 matching.

TRADES / 19

Rolls, rollover and funding survival

Goal: maintain the intended exposure as contracts, benchmarks and financing expire.

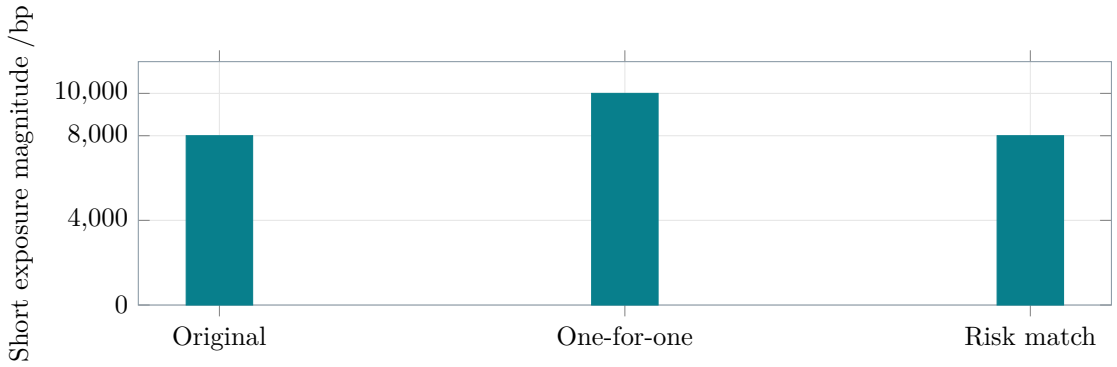
Four meanings that should not be confused

Rolldown: one bond ages along an assumed curve (page 10).

Futures roll: close an expiring contract and open a deferred one.

Benchmark roll: move exposure from an old issue to a new on-the-run issue.

Funding rollover: renew maturing repo/borrow; its rate or availability can change. "Rollout" is informal: ask which positions or dates are being moved.



Why equal contract counts do not preserve risk

Closing 100 short contracts with DV01 80 removes \$8,000/bp of short risk. Opening 100 at DV01 100 adds \$10,000/bp. Let n_{old}, n_{new} be short contract counts, and $DV01_{old}, DV01_{new} > 0$ their per-contract currency DV01 magnitudes. To keep scalar exposure constant,

$$n_{new} = n_{old}\frac{DV01_{old}}{DV01_{new}} = 100\frac{80}{100} = 80.$$

The 20-contract difference is a risk "tail". Still check factor exposures, CTD, basis and the calendar-spread price. A roll is a transaction with its own cost and liquidity, not free continuity.

A correct terminal thesis can fail before convergence

Haircuts require cash or equity in addition to financed collateral. On £20m collateral, raising haircut from 2% to 5% requires another

$$(.05 - .02)\,20m = £600,000$$

even if its value has not changed. Variation margin, falling collateral value, a more special short and borrow recall may create further demands simultaneously.

Use: size against both market losses and cash needs. Stress wider spreads, worse financing, delayed exit and lost borrowing together. State the maximum loss/cash demand the book can survive, and the rule for reducing risk if convergence is late.

RESEARCH / 20

Mean reversion: forecast a change

Goal: translate a wide residual into a horizon-dependent expected gain.

Let x_t be a spread/residual in bp, μ_x its equilibrium and $\kappa > 0$ its mean-reversion speed (inverse time units). Time t and horizon h use the same units (e.g. days). σ_x is shock volatility in bp per square-root time; B_t is standard Brownian motion, whose increment variance is dt. The Ornstein–Uhlenbeck (OU) model is

$$dx_t = \kappa(\mu_x - x_t)dt + \sigma_x\, dB_t.$$

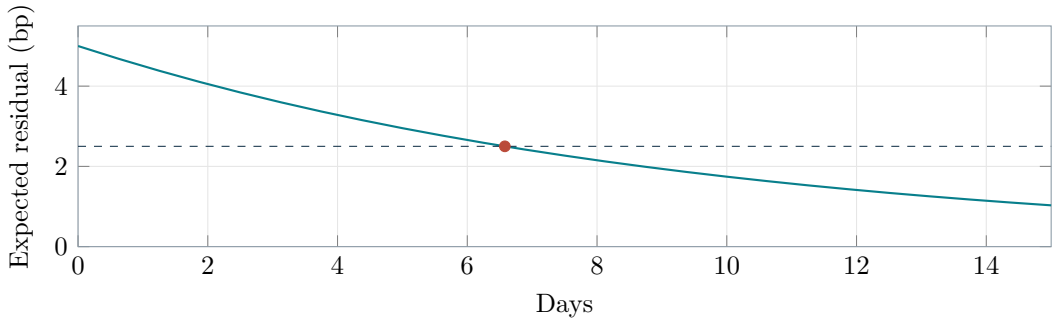
It assumes a stable equilibrium and linear restoring force, with random shocks. To solve it, set $u_t = x_t - \mu_x$ and multiply by $e^{\kappa t}$. The product rule cancels the drift:

$$d(e^{\kappa t}u_t) = \sigma_x e^{\kappa t} dB_t.$$

Integrating from t to $t+h$ gives a decaying initial condition plus a mean-zero stochastic integral. Hence

$$\mathbb{E}_t[x_{t+h}] = \mu_x + e^{-\kappa h}(x_t - \mu_x), \quad \mathrm{Var}_t(x_{t+h}) = \frac{\sigma_x^2}{2\kappa}(1 - e^{-2\kappa h}).$$

The variance follows by integrating the squared stochastic-integral coefficient. Half-life solves $e^{-\kappa h} = 1/2$: $h_{1/2} = \log 2/\kappa$.



Estimate, then use it in trade units

At regularly sampled times indexed by integer n, let Δt be the sample interval. The autoregressive model of order one (AR(1)) is $x_{n+1} = a + \phi x_n + \epsilon_{n+1}$, where a is the intercept, ϕ the persistence coefficient and ϵ_{n+1} the mean-zero innovation. The OU model implies $\phi = e^{-\kappa\Delta t}$, $a = (1-\phi)\mu_x$. For $0 < \phi < 1$, infer $\mu_x = a/(1-\phi)$ and $\kappa = -\log\phi/\Delta t$.

With $\phi = .9$, equilibrium 0 and current spread 5bp, expected five-day spread is $.9^5(5) = 2.95245$bp. Expected closure is 2.04755bp. A long-cheap position with DV01 £2,000/bp has expected gross convergence gain £4,095.10; subtract holding/execution costs.

Use: choose entry, holding horizon and size using expected closure *and uncertainty.* A z-score is $[x_t - \mu_x]/s_x$, where $s_x > 0$ is an estimated unconditional spread standard deviation. It measures unusualness, not expected profit. Estimate out of sample; quote noise, changing fair value and regime shifts can mimic or destroy mean reversion.

RESEARCH / 21

Backtests that answer a trading question

Goal: test whether the forecast can become executable net P&L.

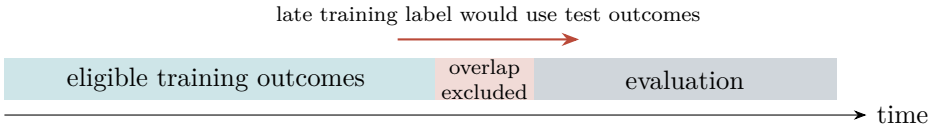
Define the target before selecting a model

Write the decision time, data actually known then, entry/exit rules, position construction and horizon. A useful target is the future change of the hedged residual, or realised net return of a fully specified trade. A good price fit alone is not evidence of forecast power.

Let x_t be the traded spread in bp, $d_t > 0$ the currency DV01 magnitude held from decision time t, and h the holding horizon. For a long-cheap spread position, net currency P&L $\Pi_{t,t+h}$ is approximately

$$\Pi_{t,t+h} \approx -d_t(x_{t+h} - x_t) + Carry_{t,t+h} - Costs_{t,t+h}.$$

The first term is convergence, the second includes financing and hedge cash flows, the last includes spread/slippage/fees. All terms use the same position and horizon.



Why chronological splitting is necessary but insufficient

An hourly label ending 24 hours later is not known until then. At each retraining time, train only on labels whose outcomes have completed. Purge training observations whose outcome windows overlap a held-out evaluation window. Fit scaling, curve choices and feature selection inside training as well.

Overlapping signals can coexist as positions; they do not create independent P&L samples. Construct one chronological portfolio path with position aggregation, turnover and capital use. Apply a realistic data-to-order delay and executable bid/ask. A touched maker price is not proof your order filled.

Try to disprove the result

Compare against carry-only and simple residual forecasts. Test wider costs, delayed execution, worse special repo, CTD changes and different regimes. Freeze specifications before a final holdout. Report the search across models, failed periods, drawdowns and capacity assumptions.

Use: distinguish an economic effect from stale quotes, unavailable information, an overfitted model or an impossible fill. The output is a reproducible net P&L path with uncertainty and failure cases, not merely a prediction score.

RESEARCH / 22

Regression, uncertainty and false reversal

Goal: know what a fitted coefficient proves and how dependent data changes confidence.

What least squares estimates

For n observations and k regressors, let $Y \in \mathbb{R}^n$ contain future spread changes, $X \in \mathbb{R}^{n\times k}$ the available predictors (including a constant if desired), and $\beta \in \mathbb{R}^k$ the unknown coefficients. Ordinary least squares (OLS) minimises $\frac{1}{2}(Y - X\beta)^\top(Y - X\beta)$. Differentiating gives

$$-X^\top(Y - X\hat{\beta}) = 0, \qquad \hat{\beta} = (X^\top X)^{-1}X^\top Y$$

when X has full column rank. The residual is orthogonal to fitted regressors in sample; this is an optimisation property, not a causal or out-of-sample claim. Use QR (orthogonal–triangular) or SVD (singular-value) decompositions rather than explicitly forming an inverse in numerical code.

Why the usual standard error can be wrong

Write $Y = X\beta + \epsilon$, where $\epsilon \in \mathbb{R}^n$ is the vector of population errors, distinct from fitted residuals $Y - X\hat{\beta}$. Conditional unbiasedness needs $\mathbb{E}[\epsilon \mid X] = 0$. Let $\Omega_\epsilon = \mathrm{Var}(\epsilon \mid X)$ be the $n \times n$ conditional error covariance. Then

$$\mathrm{Var}(\hat{\beta} \mid X) = (X^\top X)^{-1}X^\top \Omega_\epsilon X(X^\top X)^{-1}.$$

This follows by substituting $\hat{\beta}-\beta = (X^\top X)^{-1}X^\top\epsilon$. The familiar $\sigma_\epsilon^2(X^\top X)^{-1}$ assumes $\Omega_\epsilon = \sigma_\epsilon^2 I_n$, where σ_ϵ^2 is common error variance and I_n the identity matrix. Heteroskedasticity and overlapping outcomes violate it; heteroskedasticity-and-autocorrelation-consistent (HAC) or appropriately grouped/block methods estimate the relevant dependence.

Let r_t be a stationary return series, $\bar{r} = n^{-1}\sum_{t=1}^n r_t$ its sample mean, and $\gamma_k = \mathrm{Cov}(r_t, r_{t-k})$ its lag-k autocovariance. Then expand the variance of their sum and count pairs separated by k:

$$\mathrm{Var}(\bar{r}) = \frac{1}{n^2}\left[n\gamma_0 + 2\sum_{k=1}^{n-1}(n-k)\gamma_k\right].$$

Positive serial dependence means fewer effective observations. Use excess returns for a conventional Sharpe ratio. Annualising a sample Sharpe (sample mean divided by sample standard deviation) by $\sqrt{n_{yr}}$, where n_{yr} is observations per year, assumes the appropriate lack of serial dependence and a stable return process.

A mechanical source of apparent mean reversion

Suppose observed market price is $p_t = m_t + \eta_t$, where m_t is the efficient price and η_t is mean-zero quote noise. Assume efficient-price increments are uncorrelated, noise is independent through time, and noise is independent of efficient-price increments. Adjacent observed changes contain $+\eta_t$ then $-\eta_t$, so

$$\mathrm{Cov}(\Delta p_t, \Delta p_{t+1}) = -\mathrm{Var}(\eta_t).$$

A reversal coefficient may reflect bid/ask bounce that you cannot capture after paying the spread.

Use: defend confidence intervals, sampling and robustness before calling a statistical relationship alpha. More hourly rows do not cure selection bias or a broken economic link.

RESEARCH / 23

From forecasts to a risk budget

Goal: convert model scores into positions with coherent units and survivable risk.

Let $u = (u_1, \dots, u_m)^\top$ be signed position sizes in m specified tradable units (not calibration weights), π the vector of one-unit horizon P&Ls, $\mu_\pi = \mathbb{E}[\pi]$ and $\Sigma_\pi = \mathrm{Cov}(\pi)$. Then portfolio expected P&L is $u^\top \mu_\pi$, and variance is $u^\top \Sigma_\pi u$. Let $\gamma > 0$ be risk aversion, with units making return and variance terms comparable. A basic mean–variance objective is

$$\max_u \left[u^\top \mu_\pi - \frac{\gamma}{2} u^\top \Sigma_\pi u \right], \qquad \gamma > 0.$$

Differentiating gives $\mu_\pi - \gamma \Sigma_\pi u = 0$, hence $u^* = \Sigma_\pi^{-1} \mu_\pi / \gamma$ if invertible. The result says to prefer forecast return relative to joint risk, not simply the largest raw forecast.

Why units and covariance matter

Let Δx be the vector of horizon spread changes in bp, $\Sigma_x = \mathrm{Cov}(\Delta x)$, and $d_i > 0$ the currency DV01 per tradable unit of strategy i. Define $\mathsf{Q} = \mathrm{diag}(d_i)$, the DV01 scaling matrix. Each unit position profits on spread closure, so $\pi = -\mathsf{Q}\Delta x$. Hence

$$\mu_\pi = -\mathsf{Q}\mathbb{E}[\Delta x], \quad \Sigma_\pi = \mathsf{Q}\Sigma_x\mathsf{Q}, \quad v_{risk} = \mathsf{Q}u, \quad u = \mathsf{Q}^{-1} v_{risk}.$$

Here v_{risk} is signed dollar-DV01 exposure. Optimising in spread units and trading the resulting numbers as notionals is wrong. A rank score gives ordering, not expected dollars; calibrate it to a forecast or define an explicitly risk-normalised allocation rule.

Why the unconstrained solution is only a starting point

Noisy small covariance eigenvalues produce large offsetting positions. Shrink covariance, regularise exposure, and impose gross/net, factor, issuer, liquidity, turnover and funding constraints. With transaction costs, use for example

$$\max_u \; u^\top \mu_\pi - \frac{\gamma}{2} u^\top \Sigma_\pi u - \sum_i c_i |u_i - u_{old,i}|.$$

$u_{old,i}$ is the current position and c_i is currency cost per changed position unit. The absolute value discourages trades whose marginal gain cannot cover cost. Constraints may force a lower expected return to keep the book financeable.

> A trade expecting £1,200 net at a given size but losing £10,000 under a plausible spread/funding stress cannot be sized from its expected gain alone. Ask how much simultaneous loss and margin cash the portfolio can absorb, then check correlated failures and exit capacity.

Use: explain the full chain from forecast target to position units, hedge, costs, concentration and cash stress. Historical covariance is only one description of risk.

OPTIONAL BREADTH / 24

Cross-market and inflation RV

Goal: identify the replication link before comparing yields across unlike markets.

Currency hedging changes the comparison

Let spot S_{FX} and forward F_{FX} be domestic currency per foreign unit. r_d, r_f are domestic/foreign simple annual funding rates; τ is the common funding year fraction. Start with one domestic unit. Domestic investing gives $1 + r_d\tau$; converting into foreign, investing, then selling foreign forward gives $(F_{FX}/S_{FX})(1 + r_f\tau)$. Equal terminal cash flows imply covered interest parity:

$$F_{FX} = S_{FX}\frac{1 + r_d\tau}{1 + r_f\tau}.$$

Without the forward conversion, a yield comparison includes FX exposure. In practice cross-currency basis and accessible funding terms affect the hedged comparison; coupon bonds also require hedges at multiple dates.

Cross-country duration matching still leaves different policy, sovereign, liquidity and curve-factor exposures. **Use:** compare hedged cash-flow economics and residual risks, rather than raw nominal yields.

Linkers and inflation swaps: a payoff identity

To see the economic anchor, ignore lags, floors and coupons. Let I_t be the inflation index at time t, T maturity in years, and K_{inf} the annual compounded fixed inflation-swap rate. Let a real zero pay I_T/I_0 and cost P_R, while a nominal zero pays 1 and costs P_N. A zero-initial-value inflation swap receiving fixed $(1 + K_{inf})^T$ and paying I_T/I_0 converts the real-zero payoff into fixed cash. Thus

$$P_R = (1 + K_{inf})^T P_N.$$

Let y_R, y_N be real and nominal continuously compounded zero yields. Then $P_R = e^{-y_R T}$ and $P_N = e^{-y_N T}$ give

$$\log(1 + K_{inf}) = y_N - y_R.$$

This relates nominal, real and swap-implied inflation prices under the assumptions; it does not equate breakeven inflation to a physical expectation.

Why real trades deviate from the identity

Actual instruments have coupon schedules, indexation lags, seasonal inflation, floors, tax/liquidity differences and financing costs. US TIPS maturity principal is floored at original face, not at an arbitrary secondary-market purchase price. Match those features before calling a price gap RV.

Use: nominal–real–inflation-swap comparisons can reveal dislocations, but expected inflation, inflation risk premium and liquidity/funding effects are different objects. State the payoff being replicated and the unhedged risks.

Further study: be able to construct the simple replication; leave contract-specific modelling to a clearly stated extension.

OPTIONAL BREADTH / 25

Credit, options and curve shape

Goal: recognise when a smooth discount curve alone cannot explain a bond price.

Default changes which cash flows are paid

Let T be maturity, $S(t)$ the probability of surviving to t under the pricing measure, and $\rho_{rec} \in [0,1]$ recovery of face. Let $\lambda_{def}(t)$ be the risk-neutral default intensity (per year) and $D(t)$ deterministic discounting. Over dt, survival falls by $\lambda_{def}(t)S(t)dt$, giving $S(t) = e^{-\int_0^t \lambda_{def}(u)du}$. For a unit-face coupon bond with coupon rate c, accrual fractions α_j, payment times t_j, $D_j = D(t_j)$ and $S_j = S(t_j)$,

$$P = \sum_j c\alpha_j D_j S_j + D(T)S(T) + \rho_{rec}\int_0^T D(u)\lambda_{def}(u)S(u)du.$$

The terms are coupons conditional on survival, principal conditional on survival, and discounted recovery weighted by default density. This recovery convention matters; it is not equivalent to blindly adding a spread to every discount factor.

A continuously paid credit default swap (CDS) premium s_{CDS} balances protection PV:

$$s_{CDS}\int_0^T D(u)S(u)\,du = (1-\rho_{rec})\int_0^T D(u)\lambda_{def}(u)S(u)\,du.$$

Constant intensity gives $s_{CDS} = (1-\rho_{rec})\lambda_{def}$. This is a pricing relation, not an estimate of physical default probability. A bond–CDS trade retains funding, liquidity, recovery, deliverability and maturity mismatches.

Embedded options make cash flows model-dependent

Callable bonds and mortgage-backed securities (MBS) can repay early; MBS prepayments also depend on borrower behaviour. When rates fall, refinancing can shorten duration; rising rates can extend it. An option-adjusted spread (OAS) s_{OAS} solves a model valuation equation such as

$$P = \mathbb{E}^Q\left[\sum_j C_j(\text{path})e^{-\int_0^{t_j}(r_u+s_{OAS})du}\right].$$

Here Q denotes the risk-neutral pricing measure, $\mathbb{E}^Q$ its expectation, $C_j(\text{path})$ the path-dependent payment at t_j, and r_u the instantaneous short rate at integration time u. Unlike a deterministic Z-spread, this needs paths, exercise/prepayment assumptions and volatility. A large OAS can indicate model error. **Use:** compare prices under a consistent option model and hedge optionality, not just DV01.

Why parametric curves have level/slope/hump terms

Let $\beta_0, \beta_1, \beta_2$ be rate-valued shape parameters, $\tau_{NS} > 0$ a decay time in years, and $x_{NS} = T/\tau_{NS}$ dimensionless maturity. Nelson–Siegel postulates forward shape $f(T) = \beta_0 + \beta_1 e^{-x_{NS}} + \beta_2 x_{NS} e^{-x_{NS}}$. Since $z(T) = T^{-1}\int_0^T f(u)du$, integration gives

$$z(T) = \beta_0 + \beta_1\frac{1-e^{-x_{NS}}}{x_{NS}} + \beta_2\left[\frac{1-e^{-x_{NS}}}{x_{NS}} - e^{-x_{NS}}\right].$$

The last term follows by integration by parts. Short limit is $\beta_0 + \beta_1$, long limit β_0; the final loading is a hump. **Use:** obtain a low-dimensional smooth reference, while checking fit residuals, unstable parameters and whether the chosen shape distorts local RV.

PRACTICE / 26

Ten drills and project review

Commit to a sign, unit and equation before checking page 27.

1. Price 102 per 100, face £5m, modified duration 6, convexity 45. Estimate P&L for +20bp, with and without convexity.
2. A one-year zero costs 96; a two-year annual 5% bond costs 101. Find D_2 and the annual simple 1y–2y forward.
3. Reproduce the butterfly on page 8. If only 10y yield rises 4bp, what is first-order P&L?
4. Long off / short on, matched DV01 £3,000/bp. The off-minus-on spread widens 7bp. P&L? What if the short also becomes more special?
5. Haircut on £20m unchanged collateral rises from 2% to 5%. How much additional cash is required?
6. For ASW, $A = 4$, $D(T) = .82$, $c = .04$, $P_b = .97$. Find reference bond value, par swap rate and package spread.
7. Dirty price 99.5, delivery invoice 99.3, coupon 1 before delivery, $\tau = .25$, reinvestment fraction .1. Compute implied repo.
8. CTD forward DV01 72 per contract face, CF .9. Cash DV01 16,000. Futures hedge? New contract DV01 100: deferred size at roll?
9. Daily AR(1) $\phi = .95$, equilibrium 2bp, current 8bp. Find half-life and expected ten-day closure. With £2,000/bp and £1,000 drag, expected P&L?
10. Design a test of hourly 24-hour signals. Explain label availability, purging, position timing, net costs, statistical uncertainty and a falsification test.

Applied projects: three review questions

- **Funding curve:** actual instruments, economic object, equations, interpolation, solver, validation and your ownership. Do not substitute repo or credit if that is not what you calibrated.
- **XGBoost:** exact target/loss, ranks versus normalised returns, out-of-sample (OOS) dates, 24h overlap, funding, maker fills and score-to-portfolio units.
- **Production/live work:** one numerical discrepancy; one failed research idea; what your latency/speed statistic actually measures.

Five-minute RV answer: observation → economic mechanism → hedge → forecast evidence → net return → sizing/funding stress → monitoring and invalidation.

ANSWER KEY / 27

Check the reasoning, not just the number

Return to the relevant page if any sign, unit or assumption was missing.

1. $V = 5m(1.02) = 5.1m$. DV01 $= 5.1m(6)10^{-4} = 3,060$. Linear P&L $= -20(3,060) = -61,200$. Convexity $= \frac{1}{2}(5.1m)(45)(.002)^2 = 459$. Total $-60,741$ pounds.
2. $D_1 = .96$; $101 = 4.8 + 105D_2$, so $D_2 = .91619048$. Forward $= .96/D_2 - 1 = 4.78170\%$.
3. Wing notionals are $u_S = -1.40625$, $u_L = -.2109375$ million. Signed 10y DV01 $= -168.75$; P&L $= -(-168.75)(4) = +675$. Scalar level neutrality does not eliminate a twist.
4. $-3,000(7) = -21,000$. Lower special repo on the borrowed short raises its sourcing opportunity cost; funding can worsen with mark-to-market.
5. $(.05 - .02)20m = 600,000$ pounds. This cash demand exists even with unchanged collateral value.
6. $B_{ref} = .04(4) + .82 = .98$; $K_{par} = .18/4 = 4.5\%$; $s_{ASW} = (.98 - .97)/4 = .0025 = 25$bp. Check bond .97 plus swap .03 equals par.
7. $0 = 99.3 + 1(1 + .1r) - 99.5(1 + .25r)$. Thus $.8 = 24.775r$; $r = 3.22906\%$ under matched simple-rate assumptions.
8. Futures DV01 $= 72/.9 = 80$; short $16,000/80 = 200$. Roll into 160 deferred shorts at DV01 100. Recheck CTD/basis and factor exposures.
9. Half-life $= \log(.5)/\log(.95) = 13.5134$ days. Expected spread $= 2 + .95^{10}(6) = 5.59242$bp; closure $= 2.40758$bp. Net $= 2,000(2.40758) - 1,000 = 3,815.16$ pounds.
10. Only realised labels at retraining; purge overlapping outcome windows; group evaluation dates appropriately; lag to executable orders; construct one P&L path. Test financing, spread, fills, delay, regimes and selection/dependence. A favourable coefficient is not enough.

Primary references / conventions

The derivations and examples are original teaching material under stated assumptions. Selected market conventions were checked against these primary sources (17 October 2026):

- Federal Reserve: nominal yield curve methodology.
- New York Fed: repo specialness and the auction cycle.
- CME: CTD, basis, carry and implied repo; DV01 tails in futures rolls.
- US Treasury: TIPS; Bank of England: inflation-market mispricing.
- BIS: CDS–bond basis costs; New York Fed: mortgage spreads.